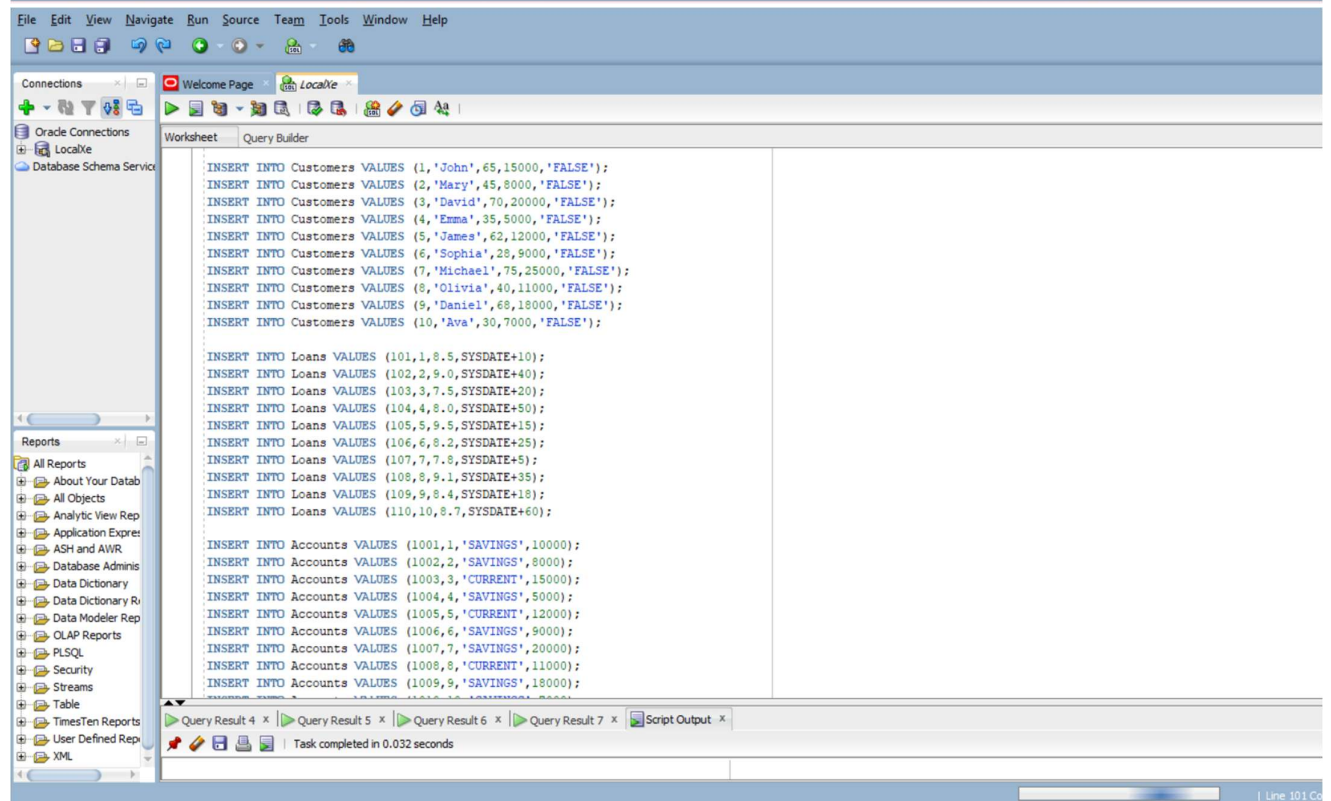
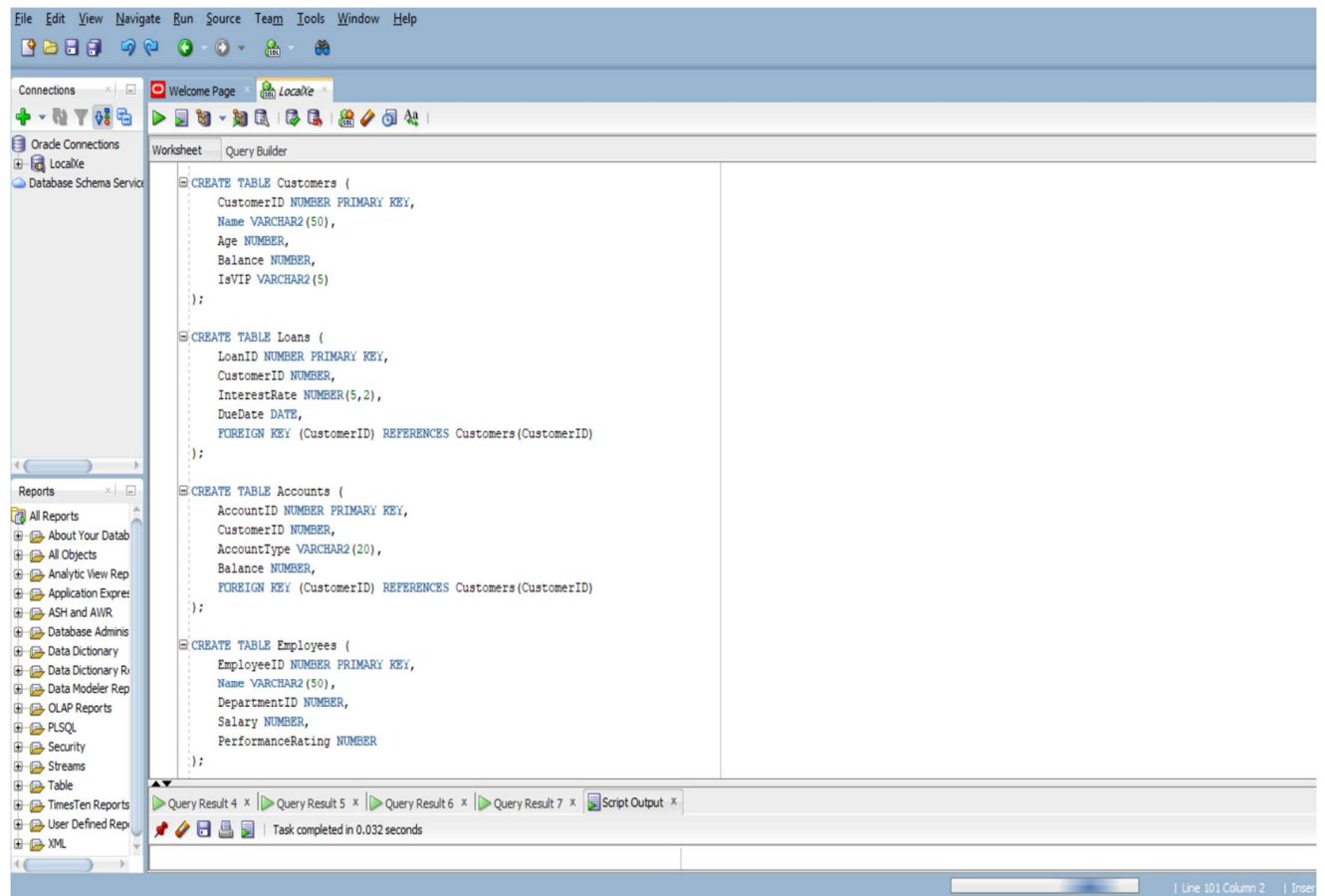


Exercise 1: Control Structures



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XML

Worksheet Query Builder

```

INSERT INTO Loans VALUES (109,9,8.4,SYSDATE+18);
INSERT INTO Loans VALUES (110,10,8.7,SYSDATE+60);

INSERT INTO Accounts VALUES (1001,1,'SAVINGS',10000);
INSERT INTO Accounts VALUES (1002,2,'SAVINGS',8000);
INSERT INTO Accounts VALUES (1003,3,'CURRENT',15000);
INSERT INTO Accounts VALUES (1004,4,'SAVINGS',5000);
INSERT INTO Accounts VALUES (1005,5,'CURRENT',12000);
INSERT INTO Accounts VALUES (1006,6,'SAVINGS',9000);
INSERT INTO Accounts VALUES (1007,7,'SAVINGS',20000);
INSERT INTO Accounts VALUES (1008,8,'CURRENT',11000);
INSERT INTO Accounts VALUES (1009,9,'SAVINGS',18000);
INSERT INTO Accounts VALUES (1010,10,'SAVINGS',7000);

INSERT INTO Employees VALUES (1,'Robert',101,50000,5);
INSERT INTO Employees VALUES (2,'Linda',101,45000,4);
INSERT INTO Employees VALUES (3,'William',102,55000,5);
INSERT INTO Employees VALUES (4,'Patricia',102,48000,3);
INSERT INTO Employees VALUES (5,'Richard',103,60000,5);
INSERT INTO Employees VALUES (6,'Barbara',103,47000,4);
INSERT INTO Employees VALUES (7,'Joseph',101,52000,5);
INSERT INTO Employees VALUES (8,'Susan',102,49000,4);
INSERT INTO Employees VALUES (9,'Thomas',103,53000,3);
INSERT INTO Employees VALUES (10,'Jessica',101,51000,5);

COMMIT;

SELECT * FROM Customers;
SELECT * FROM Loans;
SELECT * FROM Accounts;
SELECT * FROM Employees;

```

Query Result 4 x Query Result 5 x Query Result 6 x Query Result 7 x Script Output x

Task completed in 0.032 seconds

Query Result 4 x Query Result 5 x Query Result 6 x Query Result 7 x Script Output x

SQL | All Rows Fetched: 10 in 0.007 seconds

	CUSTOMERID	NAME	AGE	BALANCE	ISVIP
1	1	John	65	15000	FALSE
2	2	Mary	45	8000	FALSE
3	3	David	70	20000	FALSE
4	4	Emma	35	5000	FALSE
5	5	James	62	12000	FALSE
6	6	Sophia	28	9000	FALSE
7	7	Michael	75	25000	FALSE
8	8	Olivia	40	11000	FALSE
9	9	Daniel	68	18000	FALSE
10	10	Ava	30	7000	FALSE

Query Result 4 x Query Result 5 x Query Result 6 x Query Result 7 x Script Output x

SQL | All Rows Fetched: 10 in 0.013 seconds

	LOANID	CUSTOMERID	INTERESTRATE	DUEDATE
1	101	1	8.5	01-07-26
2	102	2	9	31-07-26
3	103	3	7.5	11-07-26
4	104	4	8	10-08-26
5	105	5	9.5	06-07-26
6	106	6	8.2	16-07-26
7	107	7	7.8	26-06-26
8	108	8	9.1	26-07-26
9	109	9	8.4	09-07-26
10	110	10	8.7	20-08-26

Query Result 4 x Query Result 5 x Query Result 6 x Query Result 7 x Script Output x

SQL | All Rows Fetched: 10 in 0.011 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE
1	1001	1	SAVINGS	10000
2	1002	2	SAVINGS	8000
3	1003	3	CURRENT	15000
4	1004	4	SAVINGS	5000
5	1005	5	CURRENT	12000
6	1006	6	SAVINGS	9000
7	1007	7	SAVINGS	20000
8	1008	8	CURRENT	11000
9	1009	9	SAVINGS	18000
10	1010	10	SAVINGS	7000

Query Result 4 x Query Result 5 x Query Result 6 x Query Result 7 x Script Output x

SQL | All Rows Fetched: 10 in 0.004 seconds

	EMPLOYEEID	NAME	DEPARTMENTID	SALARY	PERFORMANCERATING
1	1	Robert	101	50000	5
2	2	Linda	101	45000	4
3	3	William	102	55000	5
4	4	Patricia	102	48000	3
5	5	Richard	103	60000	5
6	6	Barbara	103	47000	4
7	7	Joseph	101	52000	5
8	8	Susan	102	49000	4
9	9	Thomas	103	53000	3
10	10	Jessica	101	51000	5

Scenario 1: Discount Loan Interest Rate for Customers Above 60

```
BEGIN
  FOR rec IN (
    SELECT l.LoanID
    FROM Customers c, Loans l
    WHERE c.CustomerID = l.CustomerID
    AND c.Age > 60
  )
  LOOP
    UPDATE Loans
    SET InterestRate = InterestRate - 1
    WHERE LoanID = rec.LoanID;
  END LOOP;

  COMMIT;
END;
```

Query Result 4 x | Query Result 5 x | Query Result 6 x | Query Result 7 x | Script Output x

Task completed in 0.032 seconds

PL/SQL procedure successfully completed.

	LOANID	CUSTOMERID	INTERESTRATE	DUE DATE
1	101	1	6.5	01-07-26
2	102	2	9	31-07-26
3	103	3	5.5	11-07-26
4	104	4	8	10-08-26
5	105	5	7.5	06-07-26
6	106	6	8.2	16-07-26
7	107	7	5.8	26-06-26
8	108	8	9.1	26-07-26
9	109	9	6.4	09-07-26
10	110	10	8.7	20-08-26

Scenario 2: Set VIP Status for Customers with Balance Above 10000

```
BEGIN
  FOR rec IN (
    SELECT CustomerID
    FROM Customers
    WHERE Balance > 10000
  )
  LOOP
    UPDATE Customers
    SET IsVIP = 'TRUE'
    WHERE CustomerID = rec.CustomerID;
  END LOOP;

  COMMIT;
END;
```

SSCSSCSCC

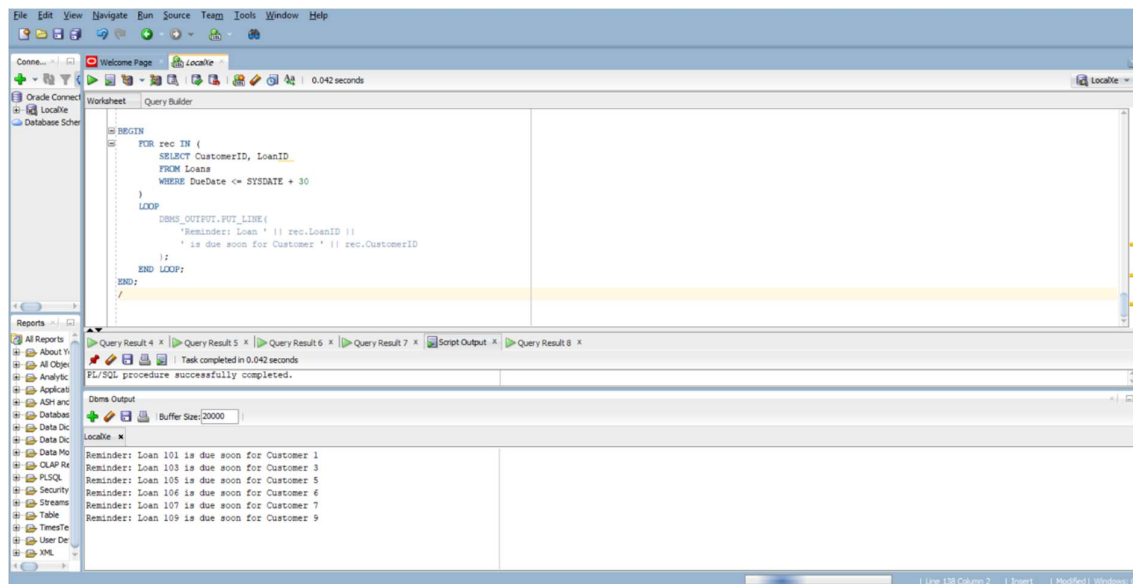
Query Result 4 x | Query Result 5 x | Query Result 6 x | Query Result 7 x | Script Output x | Query Result 8 x

All Rows Fetched: 10 in 0.008 seconds

	CUSTOMERID	NAME	AGE	BALANCE	ISVIP
1	1	John	65	15000	TRUE
2	2	Mary	45	8000	FALSE
3	3	David	70	20000	TRUE
4	4	Emma	35	5000	FALSE
5	5	James	62	12000	TRUE
6	6	Sophia	28	9000	FALSE
7	7	Michael	75	25000	TRUE
8	8	Olivia	40	11000	TRUE
9	9	Daniel	68	18000	TRUE
10	10	Ava	30	7000	FALSE

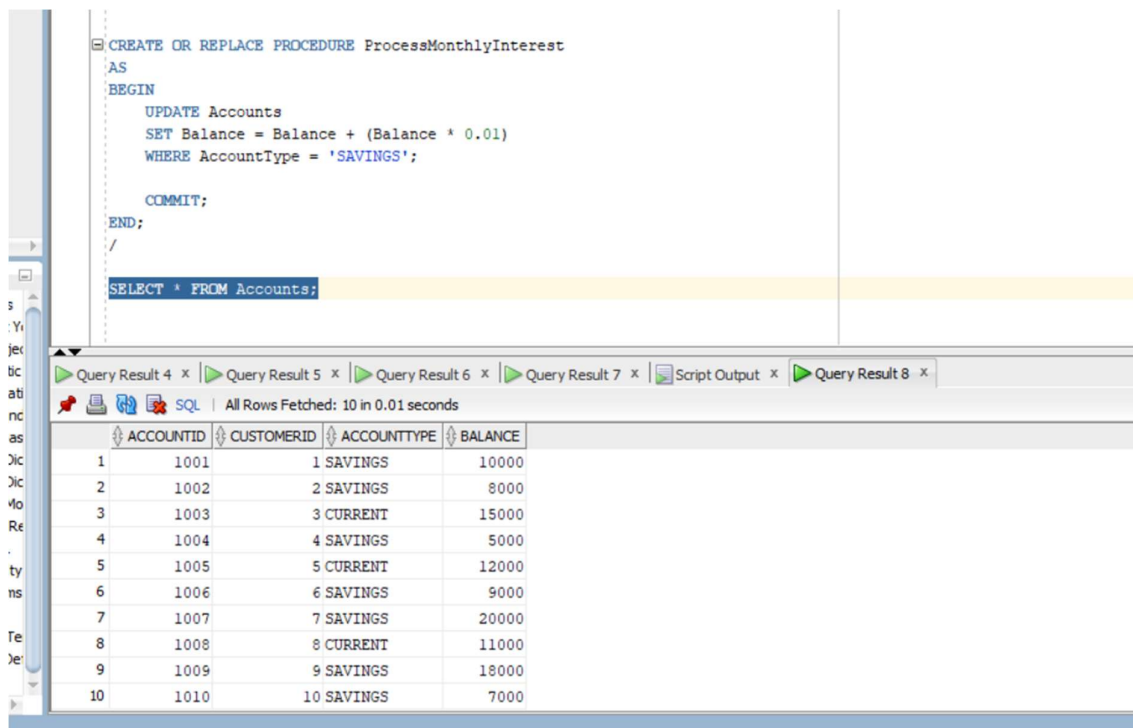
Scenario 3: Print Loan Due Reminders

```
BEGIN
  FOR rec IN (
    SELECT CustomerID, LoanID
    FROM Loans
    WHERE DueDate <= SYSDATE + 30
  )
  LOOP
    DBMS_OUTPUT.PUT_LINE(
      'Reminder: Loan ' || rec.LoanID ||
      ' is due soon for Customer ' || rec.CustomerID
    );
  END LOOP;
END;
```

Exercise 3: Stored Procedures

Scenario 1: Process Monthly Interest



Scenario 2: Set VIP Status for Customers with Balance Above 10000

```
BEGIN
FOR rec IN (
  SELECT CustomerID
  FROM Customers
  WHERE Balance > 10000
)
LOOP
  UPDATE Customers
  SET IsVIP = 'TRUE'
  WHERE CustomerID = rec.CustomerID;
END LOOP;

COMMIT;
END;
```

SQL | All Rows Fetched: 10 in 0.004 seconds

	CUSTOMERID	NAME	AGE	BALANCE	ISVIP
1	1	John	65	15000	TRUE
2	2	Mary	45	8000	FALSE
3	3	David	70	20000	TRUE
4	4	Emma	35	5000	FALSE
5	5	James	62	12000	TRUE
6	6	Sophia	28	9000	FALSE
7	7	Michael	75	25000	TRUE
8	8	Olivia	40	11000	TRUE
9	9	Daniel	68	18000	TRUE
10	10	Ava	30	7000	FALSE

Scenario 3: Print Loan Due Reminders

```
BEGIN
FOR rec IN (
  SELECT CustomerID, LoanID
  FROM Loans
  WHERE DueDate <= SYSDATE + 30
)
LOOP
  DBMS_OUTPUT.PUT_LINE(
    'Reminder: Loan ' || rec.LoanID ||
    ' is due soon for Customer ' || rec.CustomerID
  );
END LOOP;
END;
```

Task completed in 0.022 seconds

Dbms Output

Buffer Size: 20000

localXe x

Reminder: Loan 100 is due soon for Customer 6
Reminder: Loan 107 is due soon for Customer 7
Reminder: Loan 109 is due soon for Customer 9

Reminder: Loan 101 is due soon for Customer 1
Reminder: Loan 103 is due soon for Customer 3
Reminder: Loan 105 is due soon for Customer 5
Reminder: Loan 106 is due soon for Customer 6
Reminder: Loan 107 is due soon for Customer 7
Reminder: Loan 109 is due soon for Customer 9